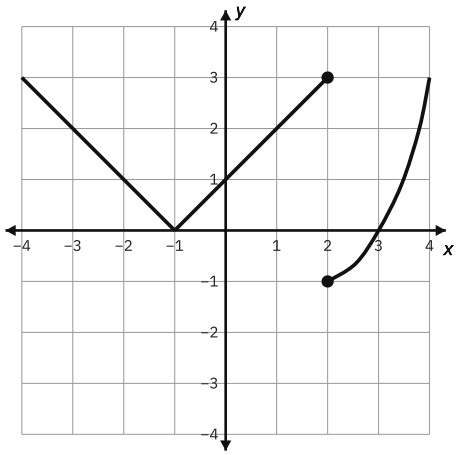
STATION

1

UNIT 5 REVIEW #2

Is it a function?

Marcel claims that the graph below represents a function.



State whether Marcel is correct. Justify your answer.

1) Yes, Marcel is correct. The graph passes the vertical line test.

2) Yes, Marcel is correct. Every *x*-value on the graph has a *y*-value.

3) No, Marcel is not correct. $x = 2$ is paired with two *y*-values, 3 and −1, so the graph fails the vertical line test.

4) No, Marcel is not correct. The graph has a V shape, so it cannot be a function.

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STATION

2

UNIT 5 REVIEW #2

Which relation is a function?

Which relation is a function?

1) {(2, 1), (3, 1), (4, 1), (5, 1)}

2) {(1, 2), (1, 3), (1, 4), (1, 5)}

3) {(2, 3), (3, 2), (4, 2), (2, 4)}

4) {(1, 6), (2, 8), (3, 9), (3, 12)}

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STATION

3

UNIT 5 REVIEW #2

Evaluate a function

If $f(x) = \dfrac{\sqrt{2x + 3}}{6x - 5}$, then $f(\dfrac{1}{2}) =$

a) 1

b) −2

c) −1

d) $-\dfrac{13}{3}$

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STATION

4

UNIT 5 REVIEW #2

Evaluate a function

A function is defined as $K(x) = 2x^2 - 5x + 3$. The value of $K(-3)$ is

a) 54

b) 36

c) 0

d) −18

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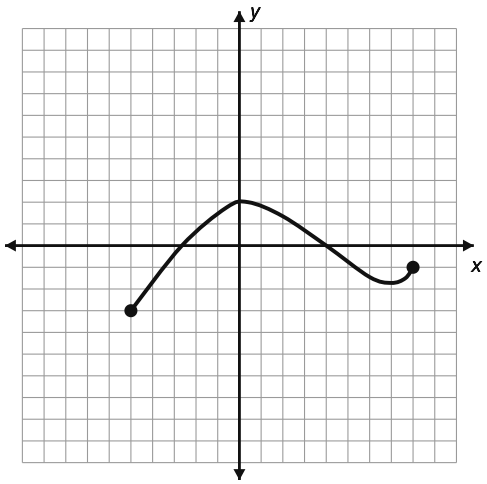
STATION

5

UNIT 5 REVIEW #2

Reading a graph

The graph below represents the function $y = f(x)$.



a) State the domain and range of this function.

A) Domain: $-5 \le x \le 8$
Range: $-3 \le y \le -1$

B) Domain: $-5 \le x \le 8$
Range: $-3 \le y \le 2$

C) Domain: $-3 \le x \le 2$
Range: $-5 \le y \le 8$

D) Domain: $-10 \le x \le 10$
Range: $-10 \le y \le 10$

b) What is the value of x when $f(x) = -3$?

c) What is $f(4)$?

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STATION

6

UNIT 5 REVIEW #2

Subtracting polynomials

What is $(x - 4)^2$ subtracted from $(x^2 + 3x - 4)$?

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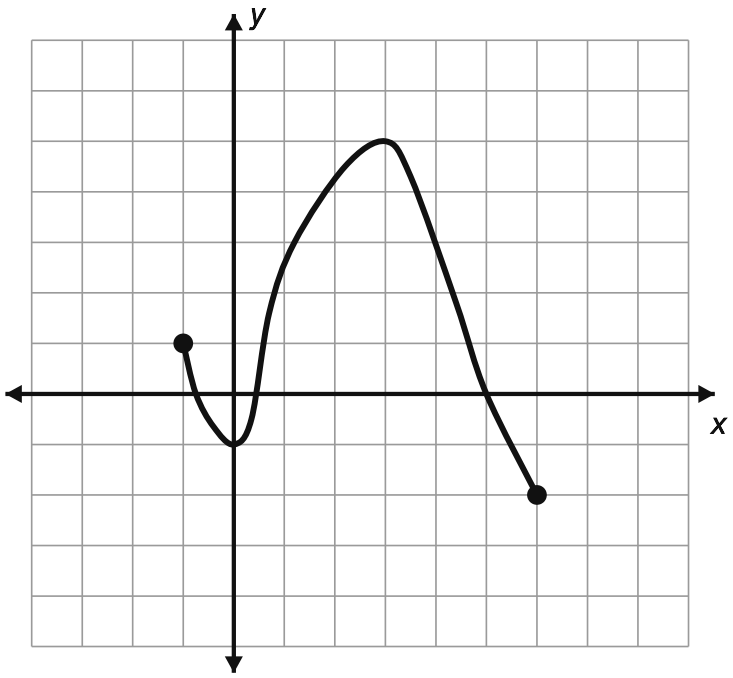
STATION

7

UNIT 5 REVIEW #2

Domain from a graph

What is the domain of the function shown below?



1) $-1 \le x \le 6$

2) $-1 \le y \le 6$

3) $-2 \le x \le 5$

4) $-2 \le y \le 5$

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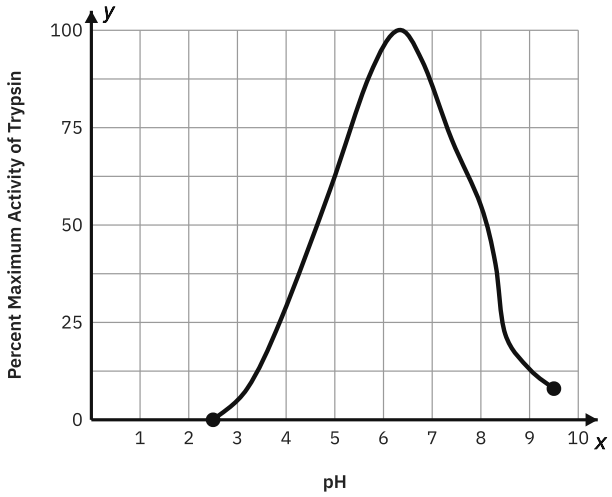
STATION

8

UNIT 5 REVIEW #2

Range of data

Data collected during an experiment are shown in the accompanying graph.



What is the range of this set of data?

1) $2.5 \leq y \leq 9.5$

2) $2.5 \leq x \leq 9.5$

3) $0 \leq y \leq 100$

4) $1 \leq x \leq 10$

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STATION

9

UNIT 5 REVIEW #2

Terms from a formula

Find the first 5 terms given the formula $f(n) = 3(n - 1) + 4$

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STATION

10

UNIT 5 REVIEW #2

Terms from a formula

Find the first 5 terms given the formula $f(n) = 2(5)^{n-1}$

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STATION

11

UNIT 5 REVIEW #2

Sequence: 1, 2, 4, 8, ...

Given the following: 1, 2, 4, 8, 16, 32, ...

a) Is the pattern Arithmetic, Geometric, or Neither?

A) Arithmetic

B) Geometric

C) Neither

b) Write a rule that represents the *n*th term of the sequence.

c) Find the 10th term of the sequence.

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STATION

12

UNIT 5 REVIEW #2

Sequence: 4, 7, 10, ...

Given the following: 4, 7, 10, 13, 16, ...

a) Is the pattern Arithmetic, Geometric, or Neither?

A) Arithmetic

B) Geometric

C) Neither

b) Write a rule that represents the *n*th term of the sequence.

c) Find the 12th term of the sequence.

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UNIT 5 REVIEW #2

Build an arithmetic sequence

Write an arithmetic sequence in which $a_1 = 3$ and $a_4 = 15$. Find the 13th term.

a) Write the sequence.

b) Find the 13th term.

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STATION

14

UNIT 5 REVIEW #2

Geometric sequence with fractions

The terms of a geometric sequence are given below.

$1,\ \frac{1}{3},\ \frac{1}{9},\ \frac{1}{27}, \ldots$

a) Write a rule for the nth term of this geometric sequence.

b) Determine the 7th term.

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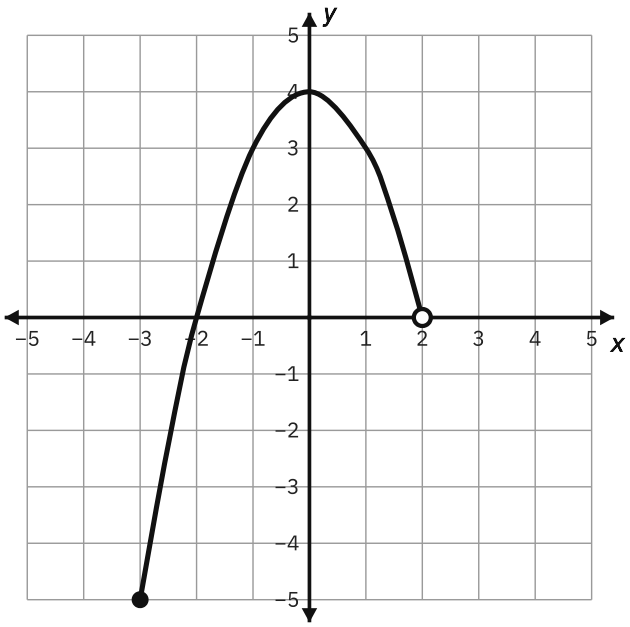
STATION

15

UNIT 5 REVIEW #2

Domain and range

State the domain and range of the function.



A) Domain: $-3 \le x \le 2$
Range: $-5 \le y \le 4$

B) Domain: $-5 \le x \le 4$
Range: $-3 \le y < 2$

C) Domain: $-3 \le x < 2$
Range: $-5 \le y \le 4$

D) Domain: $-3 \le x < 2$
Range: $-5 \le y \le 0$

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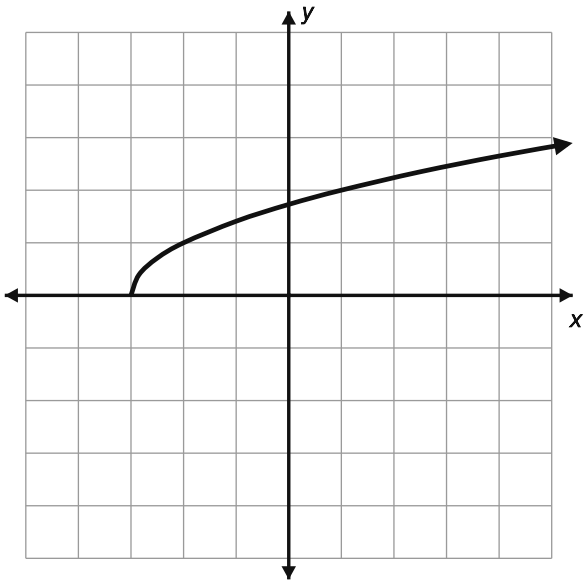
STATION

16

UNIT 5 REVIEW #2

Domain and range

State the domain and range of the function.



A) Domain: $x \geq 0$
Range: $y \geq -3$

C) Domain: all real numbers
Range: $y \geq 0$

B) Domain: $-3 \leq x \leq 5$
Range: $0 \leq y \leq 3$

D) Domain: $x \geq -3$
Range: $y \geq 0$

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STATION

17

UNIT 5 REVIEW #2

Common ratio of −2

A geometric sequence with a common ratio of −2 is

a) 6, 4, 2, 0, ...

b) 5, 7, 9, 11, ...

c) −3, 6, −12, 24, ...

d) −10, −8, −6, −4, ...

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STATION

18

UNIT 5 REVIEW #2

Solve for x

If $f(x) = \frac{5x - 9}{2}$, algebraically determine the value of x when $f(x) = 8$.

Show every step.

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